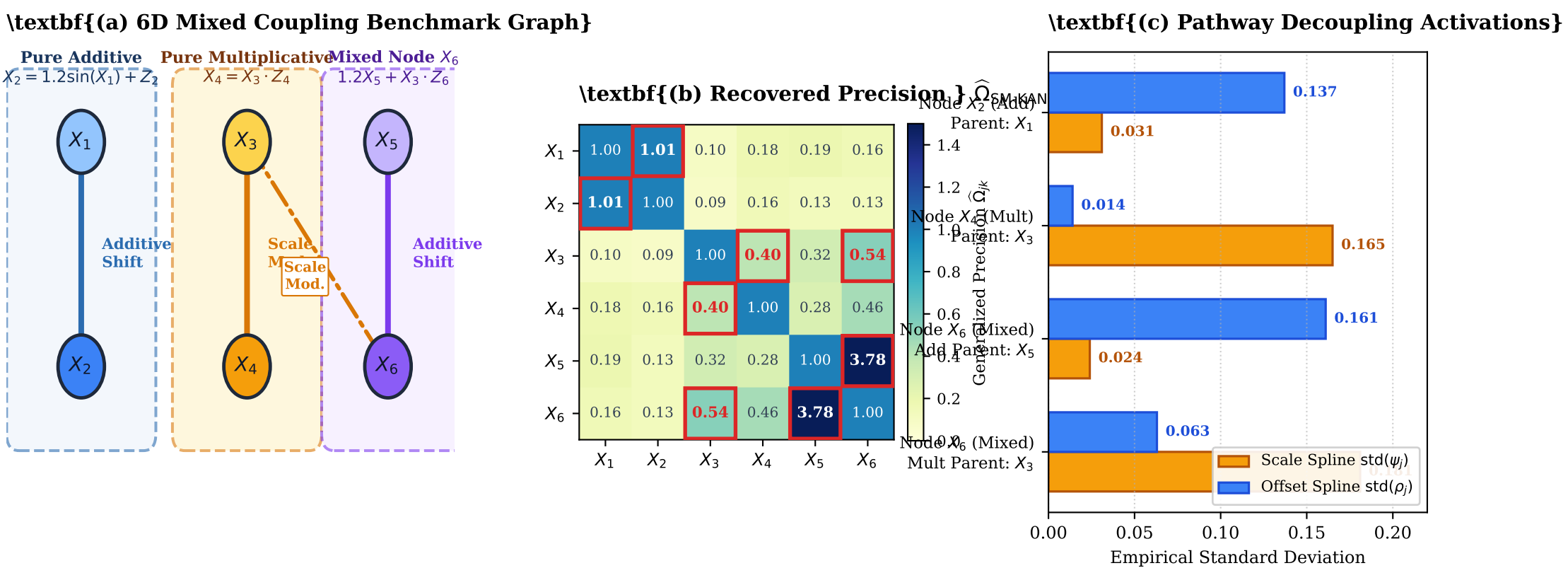

\textbf{(a) 6D Mixed Coupling Benchmark Graph}
Pure Additive
$X_2 = 1.2\sin(X_1) + Z_2$
Pure Multiplicative
$X_4 = X_3 \cdot Z_4$
Mixed Node $X_6$
$1.2X_5 + X_3 \cdot Z_6$
$X_1$
$X_2$
$X_3$
$X_4$
$X_5$
$X_6$
Additive Shift
Scale Mod.
Scale Mod.
Additive Shift
\textbf{(b) Recovered Precision } $\widehat{\Omega}_{\text{SM-KAN}}$
1.00 1.01 0.10 0.18 0.19 0.16
1.01 1.00 0.09 0.16 0.13 0.13
0.10 0.09 1.00 0.40 0.32 0.54
0.18 0.16 0.40 1.00 0.28 0.46
0.19 0.13 0.32 0.28 1.00 3.78
0.16 0.13 0.54 0.46 3.78 1.00
Generalized Precision $\widehat{\Omega}_{jk}$
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
\textbf{(c) Pathway Decoupling Activations}
Node $X_2$ (Add)
Parent: $X_1$
0.137
0.031
Node $X_4$ (Mult)
Parent: $X_3$
0.014
0.165
Node $X_6$ (Mixed)
Add Parent: $X_5$
0.161
0.024
Node $X_6$ (Mixed)
Mult Parent: $X_3$
0.063
Scale Spline std($\psi_j$)
Offset Spline std($\rho_j$)
0.00
0.05
0.10
0.15
0.20
Empirical Standard Deviation